DeepSkin

INFO9023 - Spring 2025

Created By

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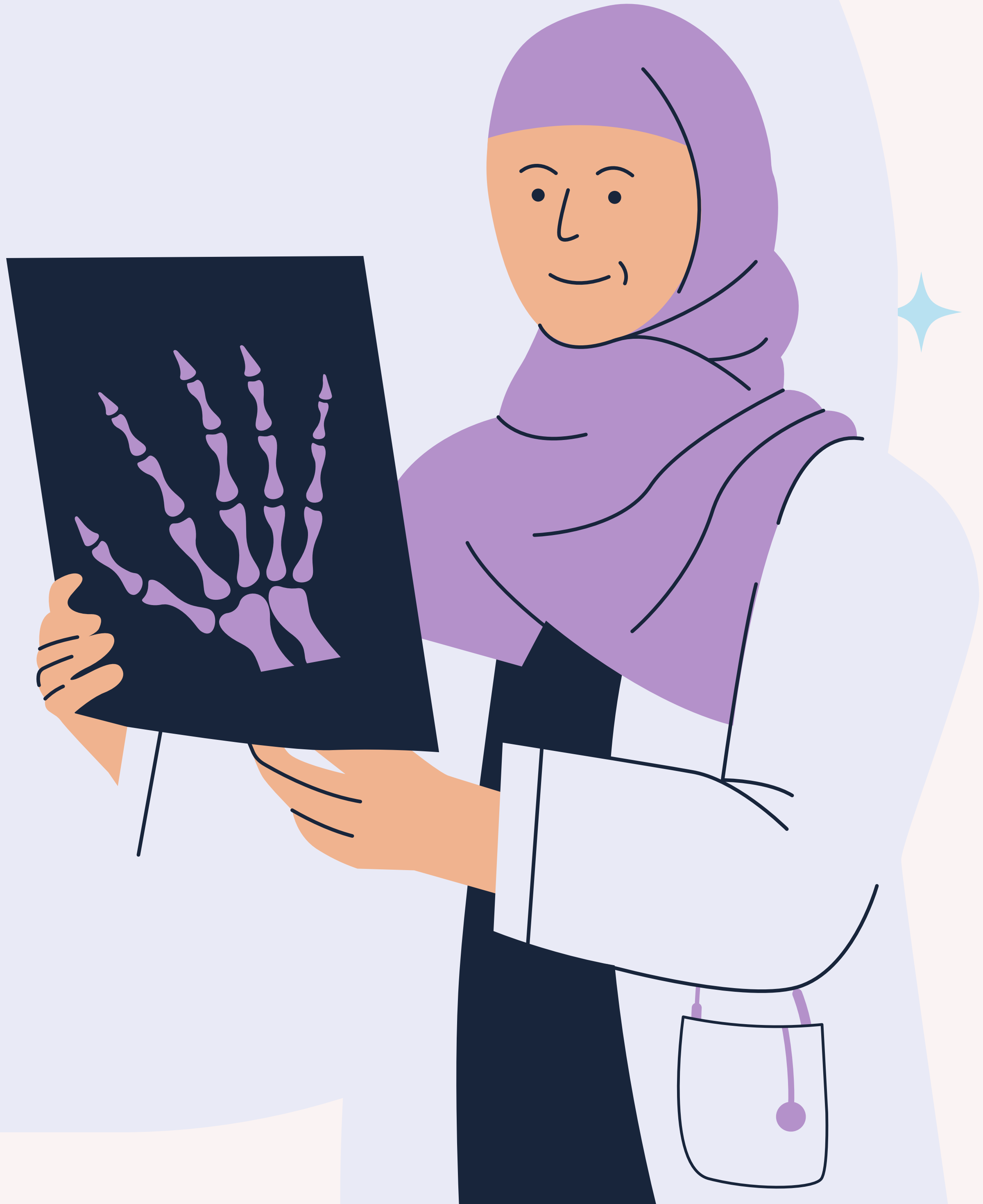
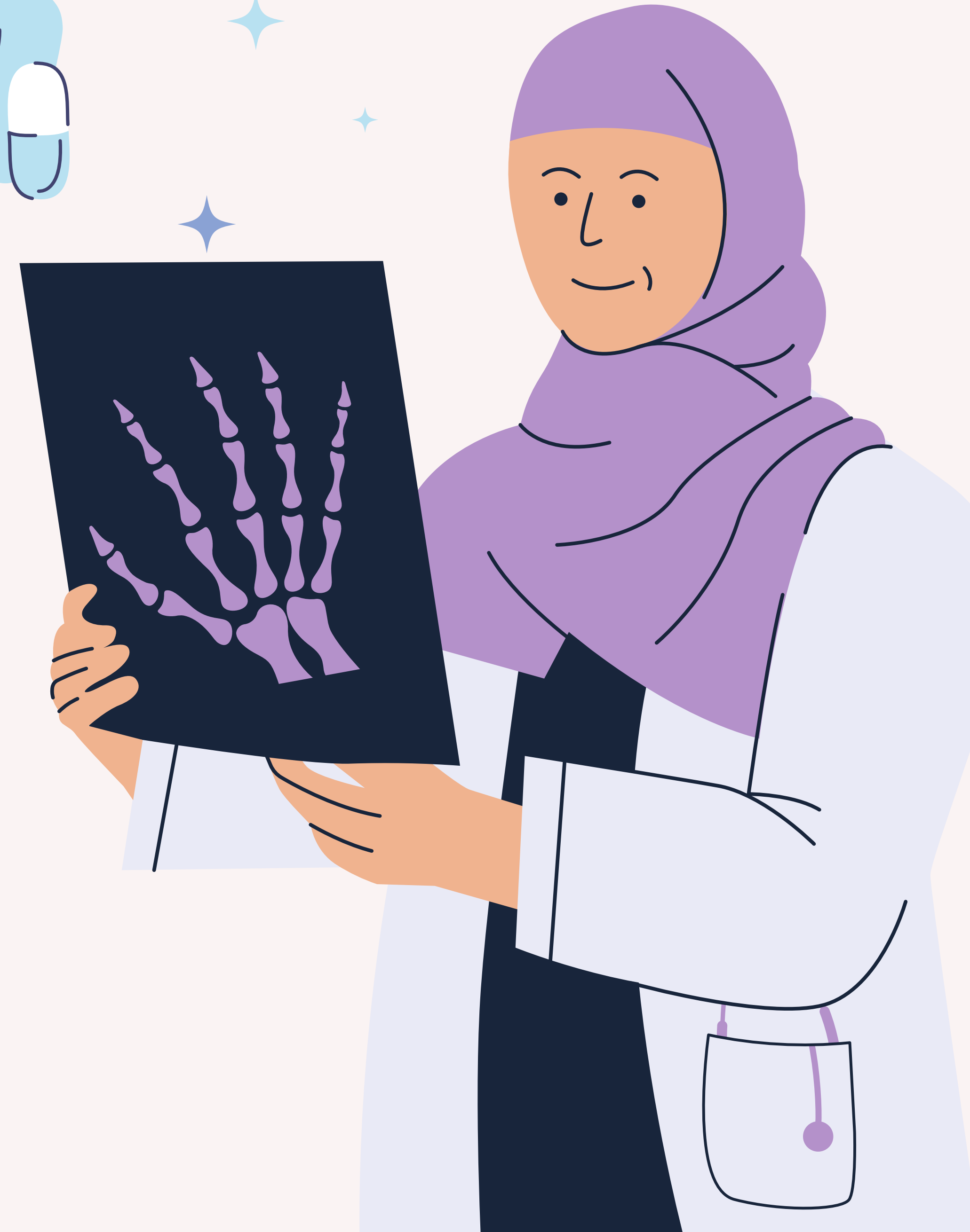


Project Reminder



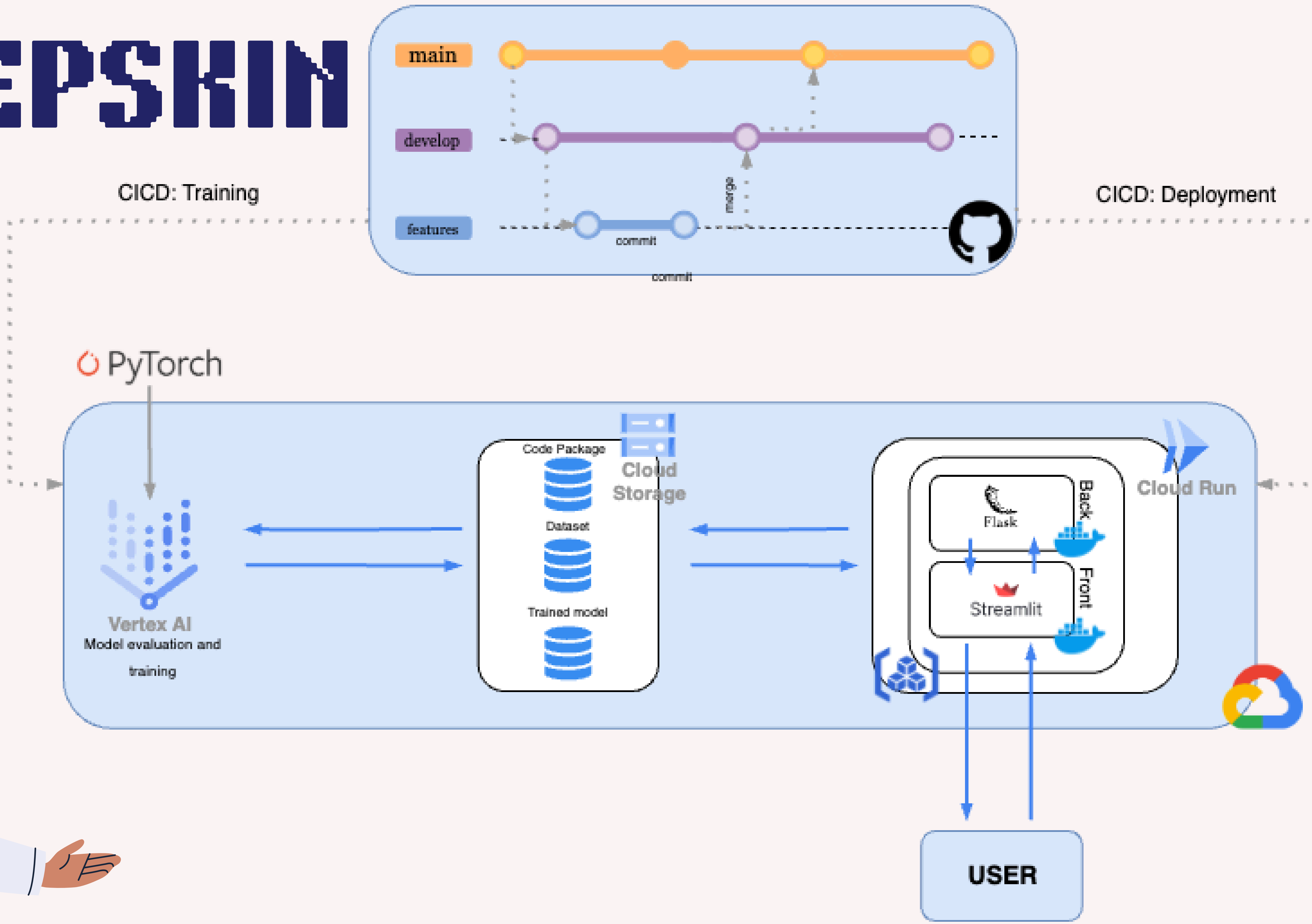
DEEPSKIN

Skin cancer is one of the most common forms of cancer, and early detection is crucial for effective treatment. DeepSkin aims to bridge this gap by providing an AI-powered solution for analyzing skin lesions.

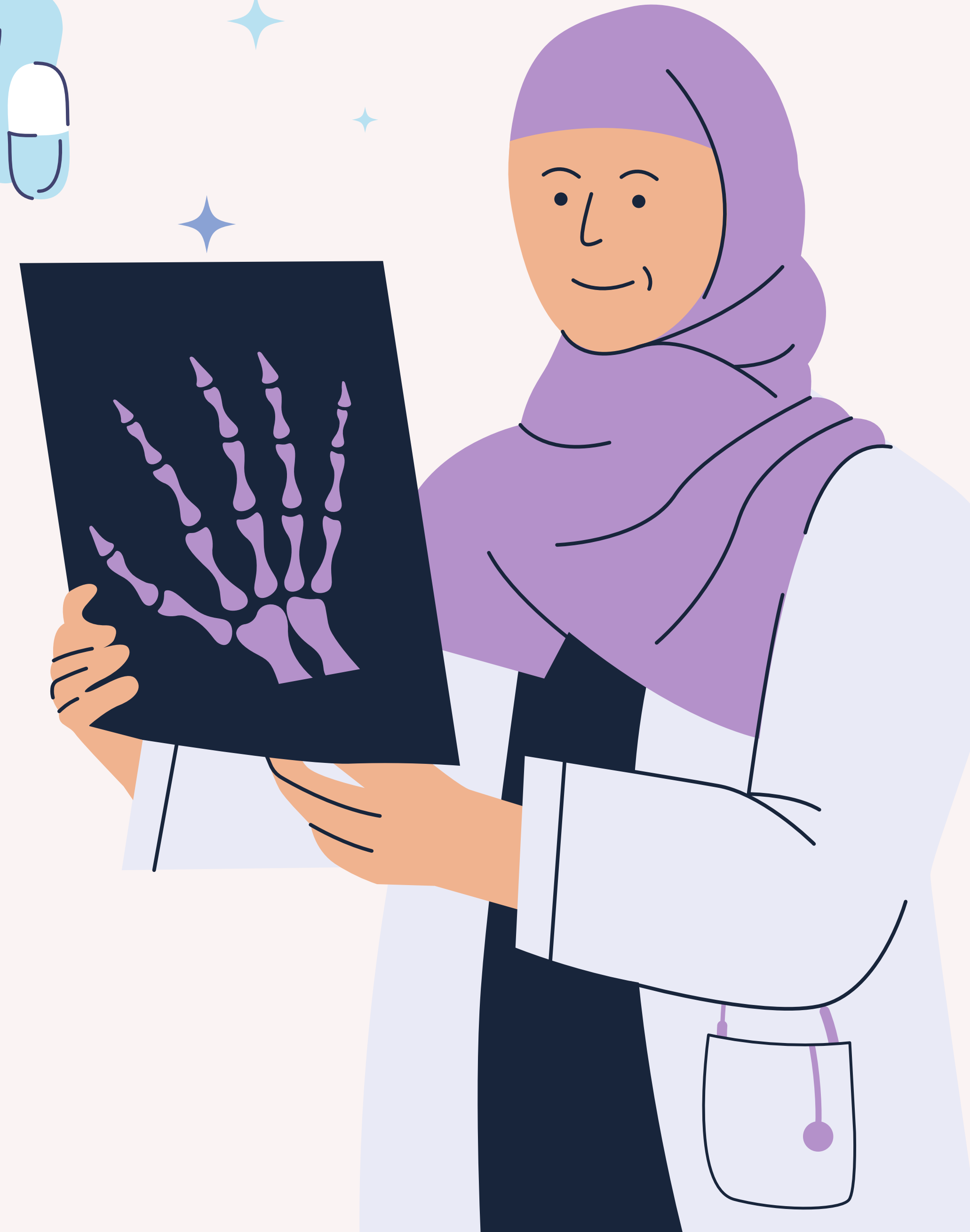
Users can upload images of their skin lesions, and our deep learning model will assess whether the lesion is likely to be malignant or benign.

Global Scheme

DEEPSKIN


main
develop
features
commit
commit
merge
CICD: Training
CICD: Deployment
PyTorch
Vertex AI
Model evaluation and training
Code Package
Dataset
Trained model
Cloud Storage
Flask
Back
Streamlit
Front
Cloud Run
USER

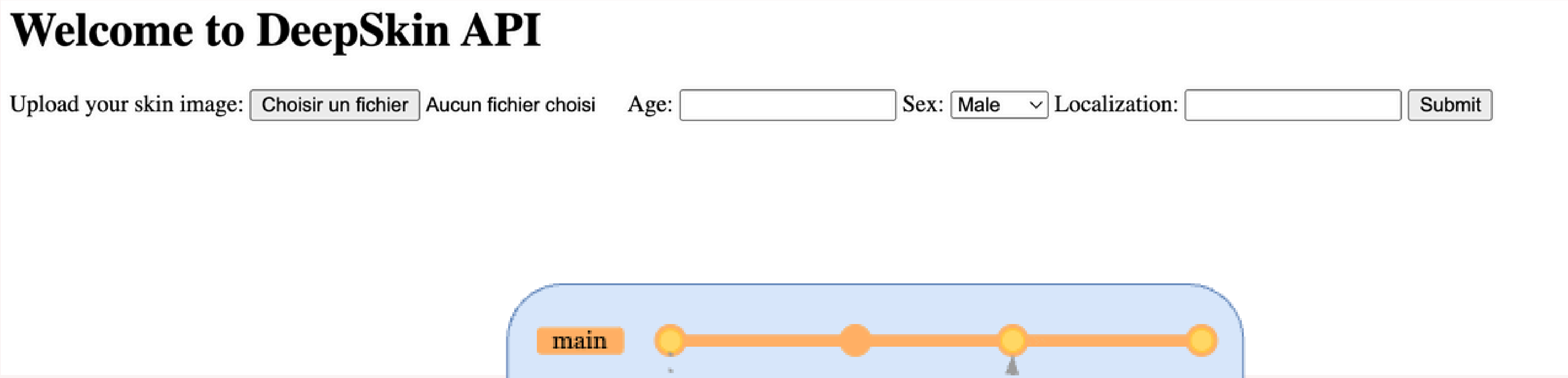
Deployment



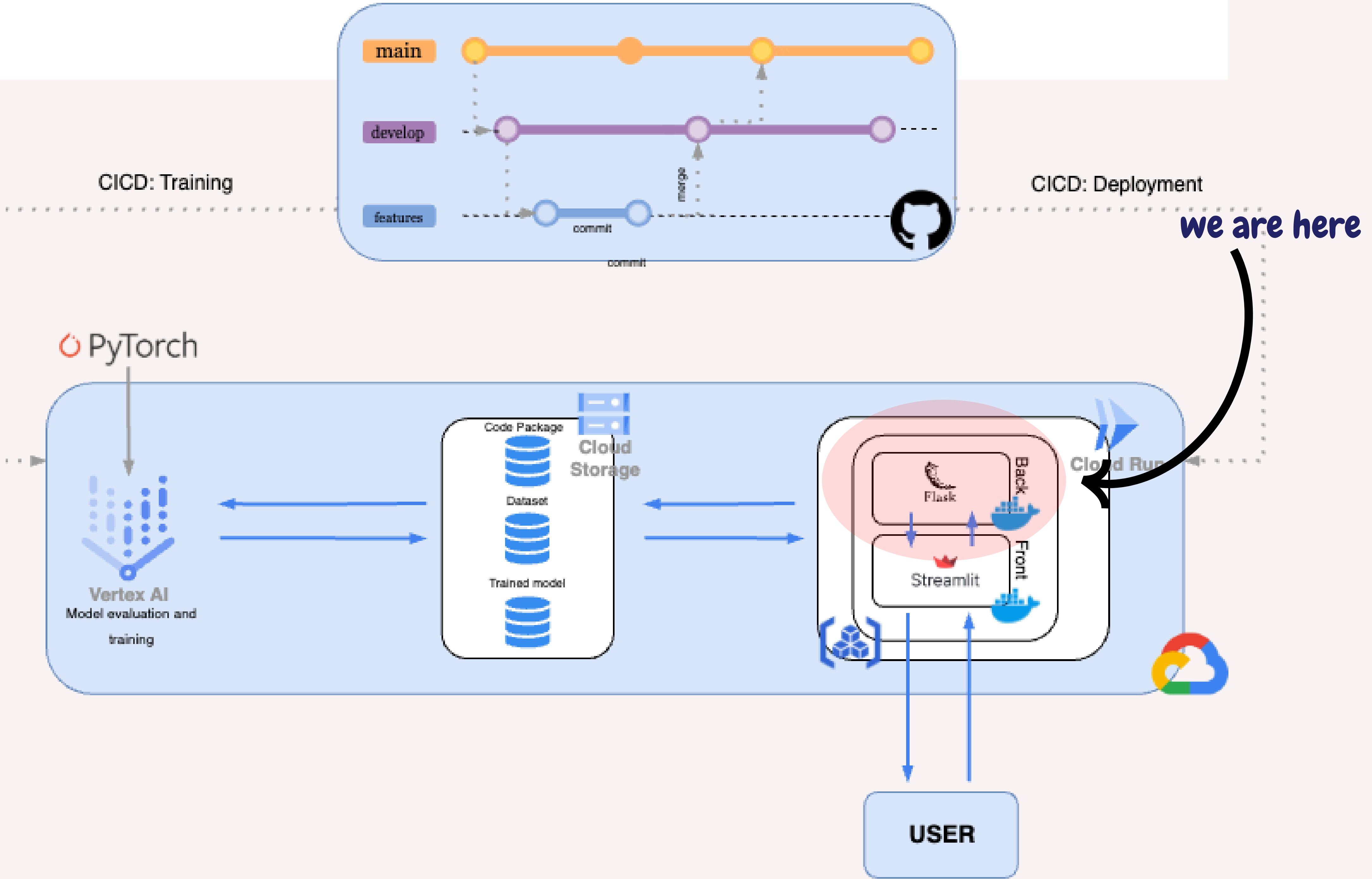
DEPLOYMENT

Backend Deployment

The **backend** of our application is made using flash. We define a root route and a predict route. The deployment was at first done **locally** using **Docker** we then push it in Google Could **Artifact** and run it on Google Cloud **Run**


Welcome to DeepSkin API

Upload your skin image: Choisir un fichier Aucun fichier choisi Age: Sex: Male Localization: Submit



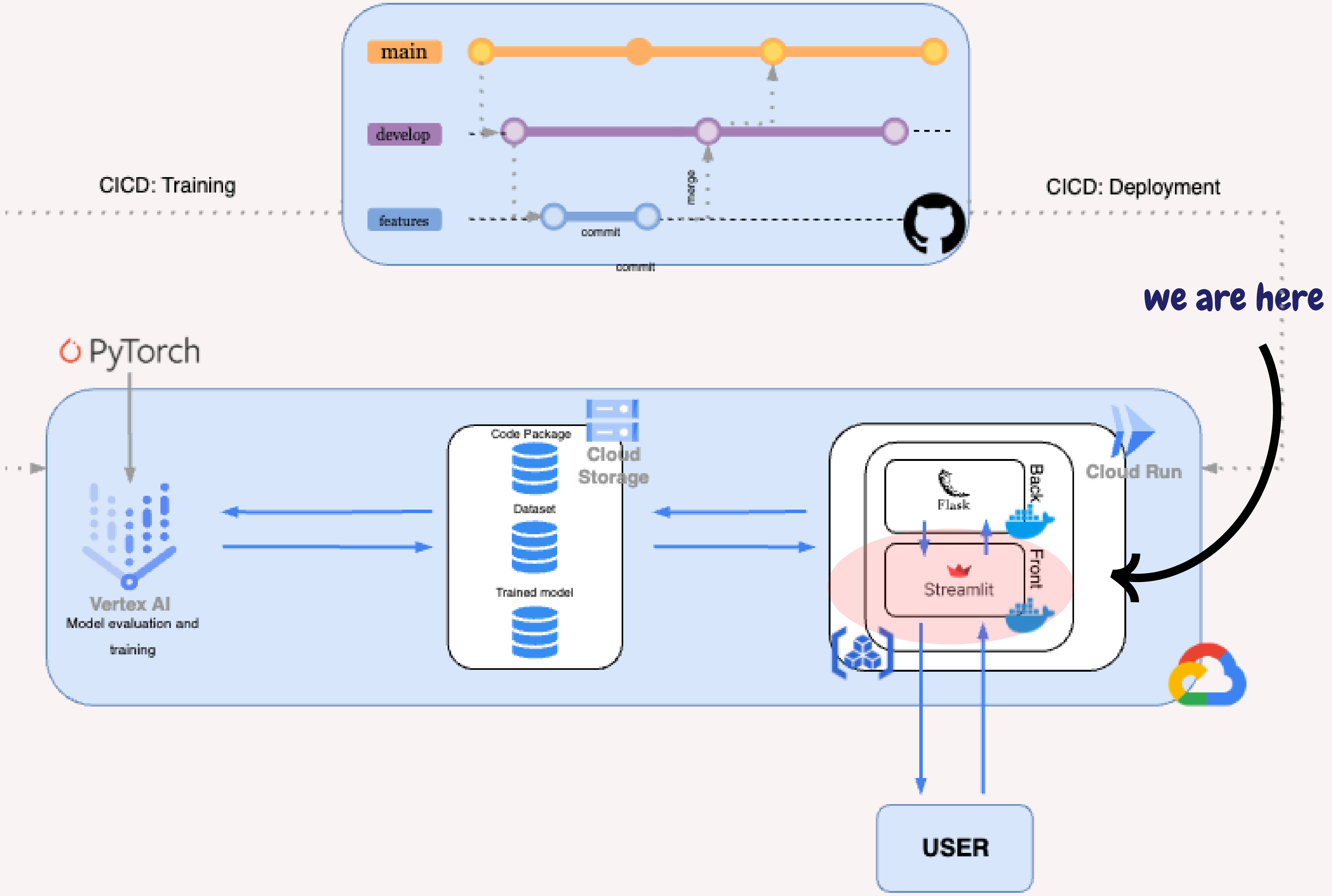
Deployment

DEPLOYMENT

Frontend Deployment & Monitoring

For the frontend we used Streamlit. We did that to have a full python solution for our application in which no front-end experience is required as making clean UI is a complex task. Now let's show you a quick demo of the app and the dashboard we created

https://deepskin-front-808565425437.europe-west1.run.app

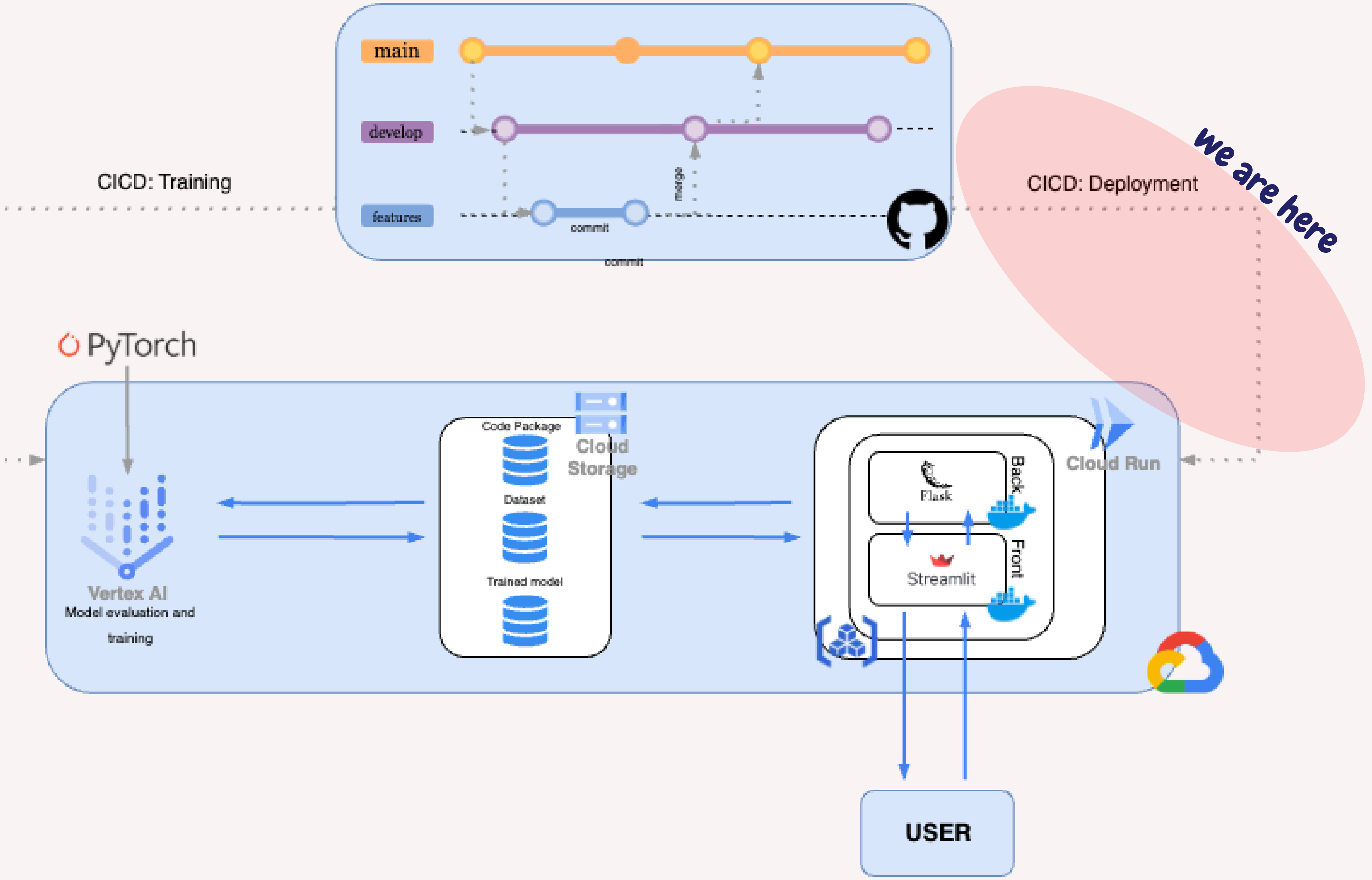


Deployment

DEPLOYMENT

CICD: Automatic Deployment

Using Github Actions and Secrets we were able to create CICD script for automatic deployment. This automatic script is trigger when a push is made on a branch and it will redeploy automatically both back and front-end

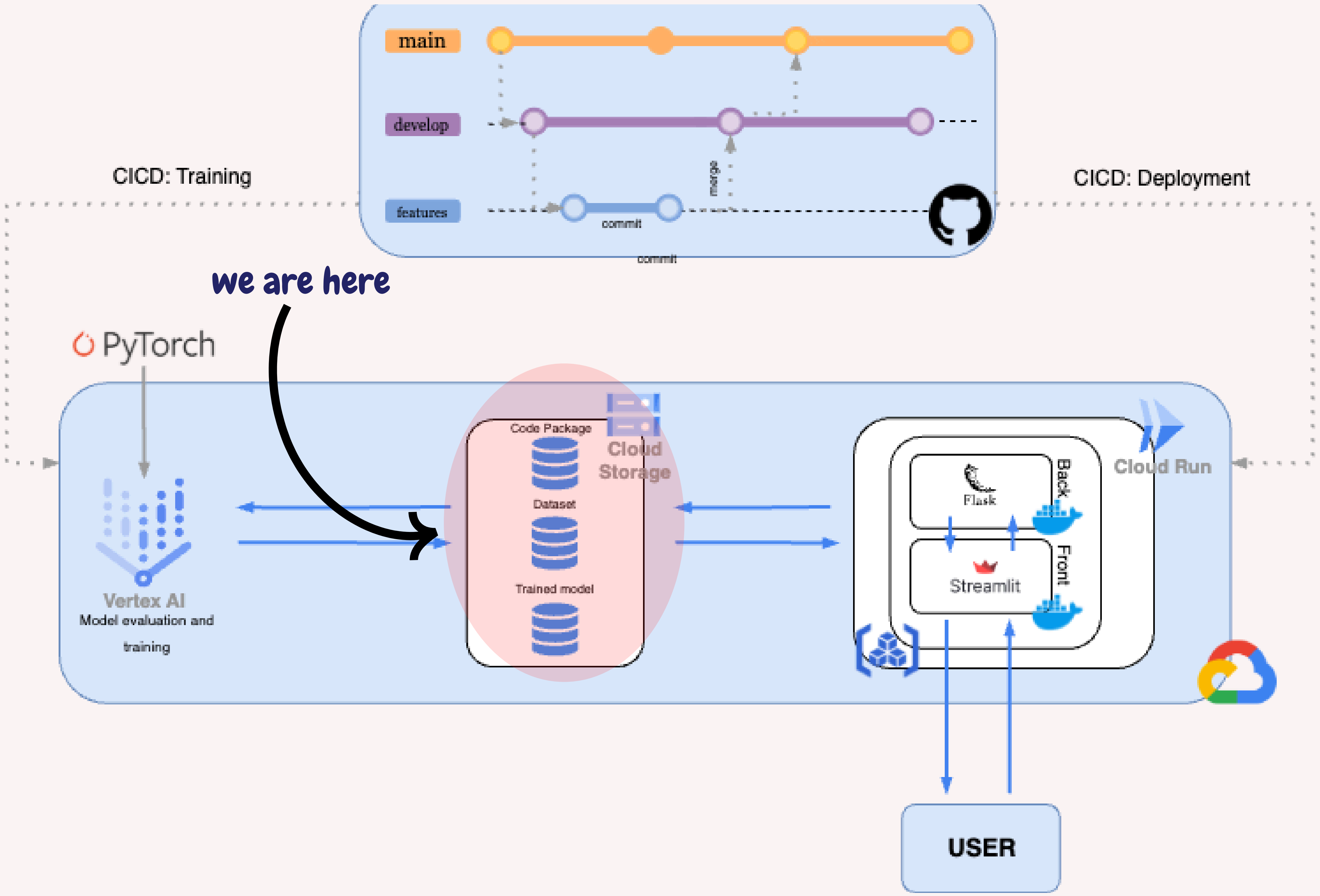


Training

Training

Storage structure

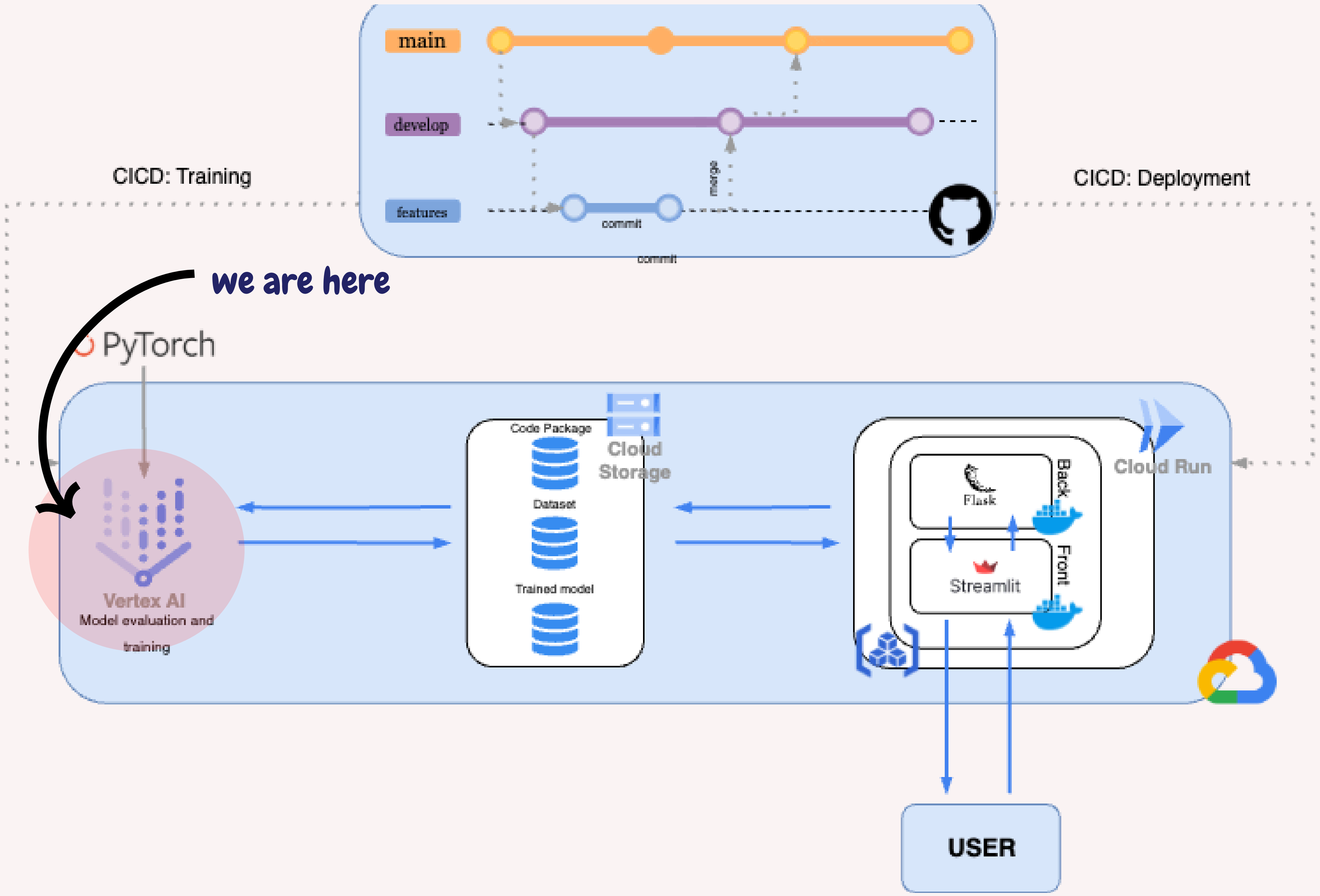
The overall structure of our application is centered around 3 buckets we created inside GCloud Storage. They respectively contains: our codebase, our dataset and finally the final trained model.



Training

Vertex IA

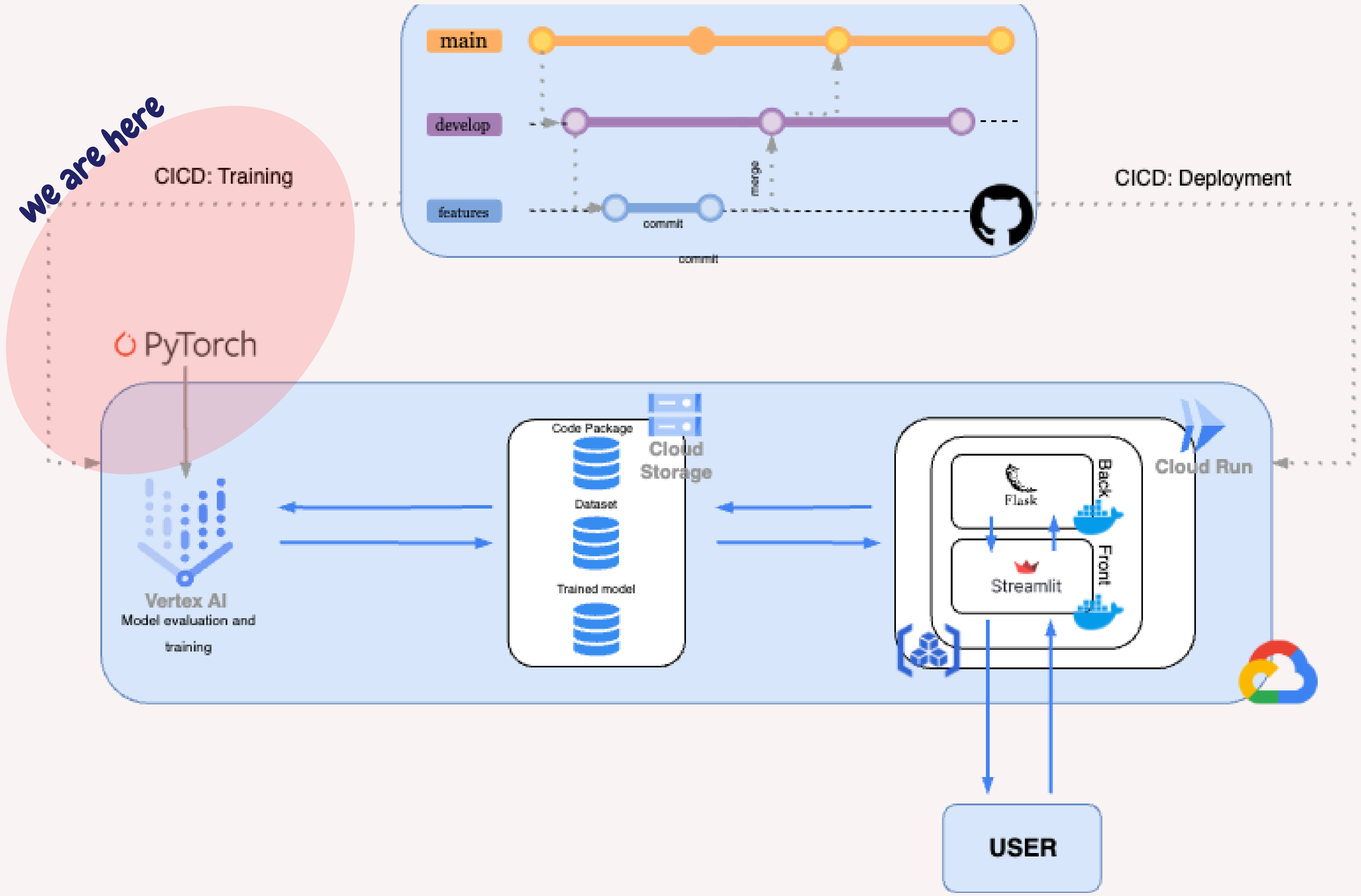
To train our model in the cloud we used vertex IA. This approach uses of a pre-built PyTorch image to not have to create a huge docker image. Vertex IA requires the codebase to be inside a python package so we encapsulate it using a new setup.py script



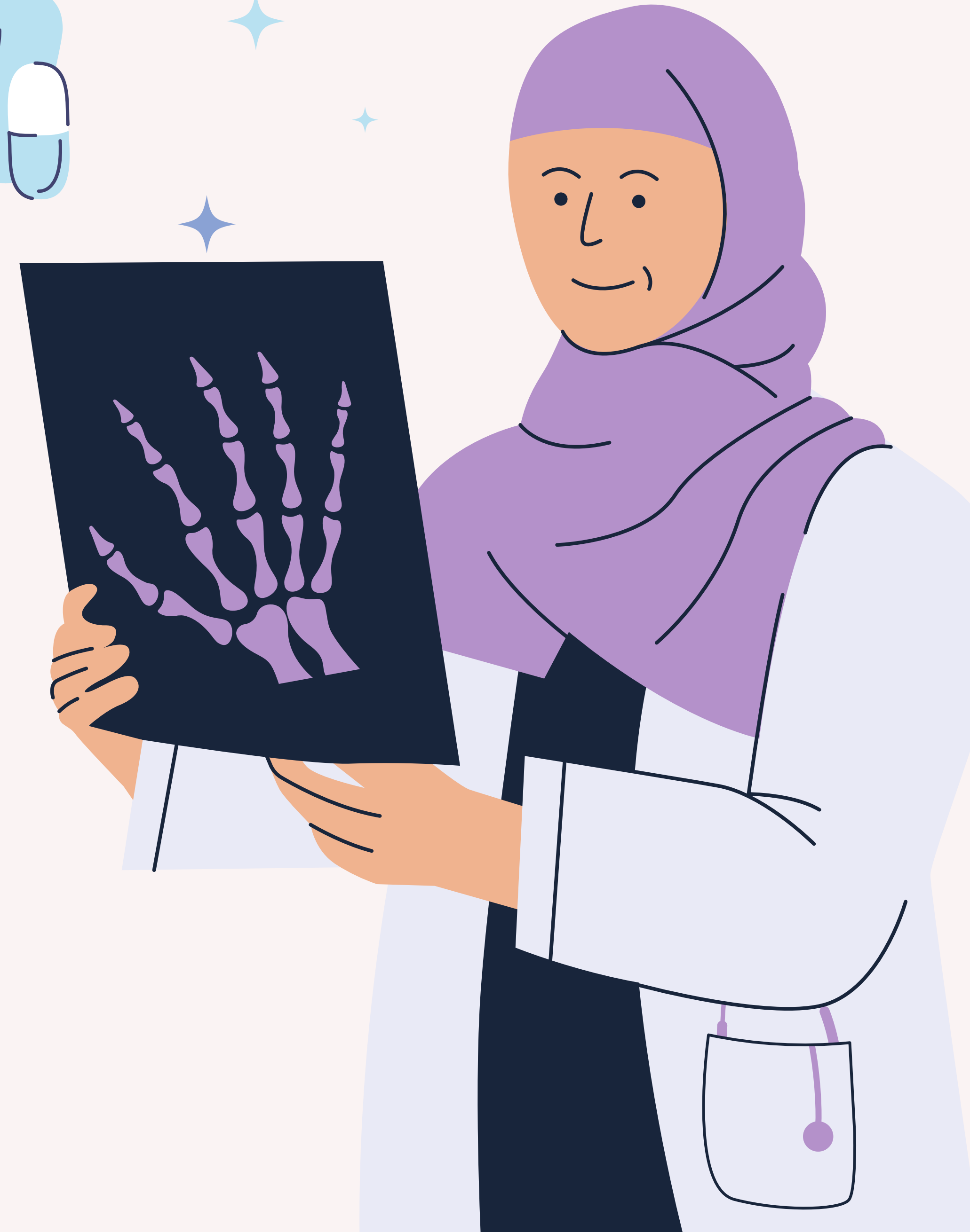
Training

CICD: Automatic Training

Using Github Actions and Secrets we were able to create CICD script for automatic training. This automatic script is trigger when a special message is incorporated in the commit message (!train-vertex!)



Complete Overview

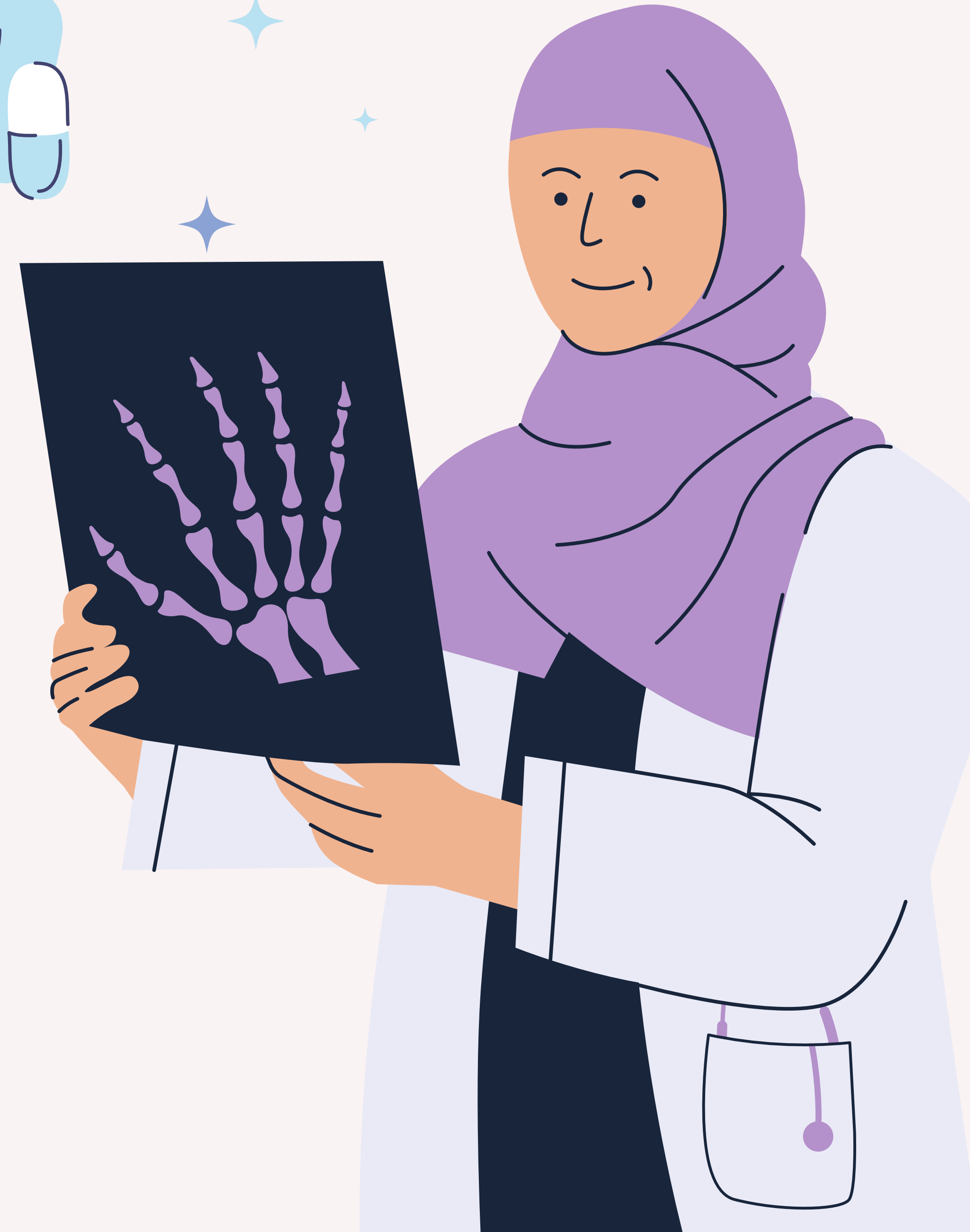


Overview

Regarding the current state of DeepSkin:

- We implemented CICD actions to fully automate parts of the application such as training and deployment
- We started testing and checking our code to ensure every part of the codebase was working as expected such as the API,...
- We leverage the efficiency of Gpu and divide our training time by 5
- We improved our documentation

What's Next ?



Our Next Step / Improvements

Bash Script

We will simplify our life by creating batch script to pass Hyperparameters value directly on the run command

More Documentation

Currently our application lacks lots of explication and examples we will add more documentation

More Test

We will add more tests all around our application to ensure that everything is working perfectly

WandB Monitoring

We will also integrate WandB into our codebase to monitore the performances of our model as well as adding more metrics to access its quality

Thank you for your attention

Any Questions ?